

SENTINEL AI — DEEPPFAKE INVESTIGATION REPORT

INVESTIGATION ID: CASE-TEST-M7 — DIGITAL EVIDENCE EXAMINATION

1. INVESTIGATION INFORMATION

|                 |                                       |                   |                              |
|-----------------|---------------------------------------|-------------------|------------------------------|
| Case ID:        | CASE-TEST-M7                          | Case Status:      | Under Investigation          |
| Case Title:     | Milestone 7 Frontend Integration Test | Report Generated: | 01 October 2026, 06:06 UTC   |
| Investigator:   | Test User                             | Investigator ID:  | INV-001                      |
| Case Submitter: | Test User                             | Submitter ID:     | USR-001                      |
| Incident Date:  | N/A                                   | Analysis Engine:  | Sentinel AI V1.7-A Dual-Head |

2. EVIDENCE SUMMARY

| Category               | Count | AI Forensic Classification | Count |
|------------------------|-------|----------------------------|-------|
| Total Evidence:        | 1     | AI Scanned:                | 1     |
| User Uploaded:         | 1     | Authentic:                 | 0     |
| Investigator Uploaded: | 0     | Tampered:                  | 1     |

3. DETAILED EVIDENCE EXAMINATION

EVIDENCE ITEM 1 (EV-00014) — target\_m7\_photo.jpg

|                    |                                                                         |             |                      |
|--------------------|-------------------------------------------------------------------------|-------------|----------------------|
| Evidence ID:       | EV-00014                                                                | AI Scan:    | Completed            |
| Original Filename: | target_m7_photo.jpg                                                     | Prediction: | TAMPERED             |
| Uploaded By:       | Test User                                                               | Confidence: | 75.01%               |
| Uploader Role:     | Investigator                                                            | Model:      | V1.7-A               |
| Uploaded At:       | 24 September 2026, 21:45 UTC                                            | Scanned At: | 2026-10-01 06:06 UTC |
| File Size:         | 41.20 KB                                                                | MIME Type:  | image/jpeg           |
| SHA-256 Hash:      | 57244f424f284ffbb6aae62425f06ea500000<br>000000000000000000000000000000 |             |                      |



Figure 1: Original Evidence Image



Figure 2: Sentinel AI Heatmap Overlay

| Metric                | Sentinel AI V1.7-A Value     | Threshold Criteria     |
|-----------------------|------------------------------|------------------------|
| Classification Result | TAMPERED / MANIPULATED MEDIA | Threshold: 0.40        |
| Tampered Probability  | 75.01%                       | Range: [0.0% - 100.0%] |
| Authentic Probability | 24.99%                       | Range: [0.0% - 100.0%] |
| Spatial Localization  | Heatmap Artifact Generated   | Threshold: 0.35        |

## 4. MODEL METHODOLOGY & TECHNICAL DETAILS

**Architecture Overview:** Sentinel AI V1.7-A is a frozen dual-head deep learning architecture optimized for image deepfake detection and pixel-level spatial localization.

**Preprocessing Stream:** Input images are normalized and passed through a dual-stream RGB backbone (EfficientNet-B0) paired with a high-pass Laplacian residual filter bank capturing high-frequency forensic noise signatures.

**Dual Heads:**

- *Classification Head:* Computes softmax probability across authentic vs. tampered classes using a calibrated decision threshold of **0.40**.
- *Localization Head:* Generates pixel-level feature heatmaps highlighting manipulated regions using a spatial activation threshold of **0.35**.

## 5. FORENSIC INTERPRETATION & DISCLAIMER

**OFFICIAL FORENSIC DISCLAIMER**

This report presents automated deepfake classification and spatial localization results produced by Sentinel AI V1.7-A. These findings provide probabilistic indicators based on forensic noise, boundary inconsistencies, and facial feature distributions. This document is intended to assist law enforcement, legal experts, and digital forensic investigators as corroborative evidence. Final evidentiary conclusions should incorporate qualified human expert examination.